

# **Final Assignment**

## **Disease Prediction Using Machine Learning Regression Models**

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## Abstract

The severity of diseases prediction is an important tool to enhance the decision-making process and early intervention in healthcare. The proposed project is aimed at forecasting the severity of a disease based on supervised machine learning regression methods on a disease prediction dataset in the real world. The correct prediction of severity might also help health care workers to prioritize treatment and allocate resources effectively.

The data is comprised of symptoms of patients and medical factors that have an impact on the severity of the disease. This research is in line with the United Nations 2030 sustainable development goal 3 (Good Health and Well-being), and evidence-based healthcare interventions are appropriate to enhance the health results of the population. The project uses end-to-end machine learning workflow, such as exploratory data analysis, preprocessing, regression model development, hyperparameter optimization and feature selection. The classical regression models and a neural network regressor were evaluated based on MAE, RMSE and R2 measures.

## Introduction

### Problem Statement

The levels of disease occurrence in patients differ greatly due to the symptoms, physiological nature, and underlying factors. It is crucial to predict the severity of the disease in advance to provide the opportunity to respond to it in time and plan proper treatment. Assessment may be subjective and time-consuming, which explains why machine learning models are encouraged in the process of automated severity prediction.

The formulation of this task as a regression problem is based on the fact that the target variable is an ordinal or a continuous severity score.

### Dataset Description

The data set utilized in this project will have medical and symptom attributes of the patients. These characteristics are quantifiable factors leading to severity of the disease. The dataset is real-life healthcare and consists of numerical variables that can be regressed.

This project promotes UNSDG 3, because predictive healthcare analytics increase efficiency of preventive care and treatment.

### Objectives

These regression task objectives are:

- To investigate and analyze the disease prediction data.
- To do exploratory data analysis and trends.
- To pre-process data on regression modelling.
- To apply classical and neural network regression.
- To use hyperparameter optimization and feature selection.
- In order to contrast models and find the most effective way of doing it.

## Exploratory Descriptive Statistics (EDA)

The purpose of EDA was to get an idea about the feature distributions, outliers, and the correlation between variables. Missing data and inconsistencies were verified in the dataset prior to developing the model.

Figure 1: Target Variable Distribution (Histogram).

This histogram demonstrates the prevalence of the levels of disease severity. It shows the distribution of the severity values among the patients.

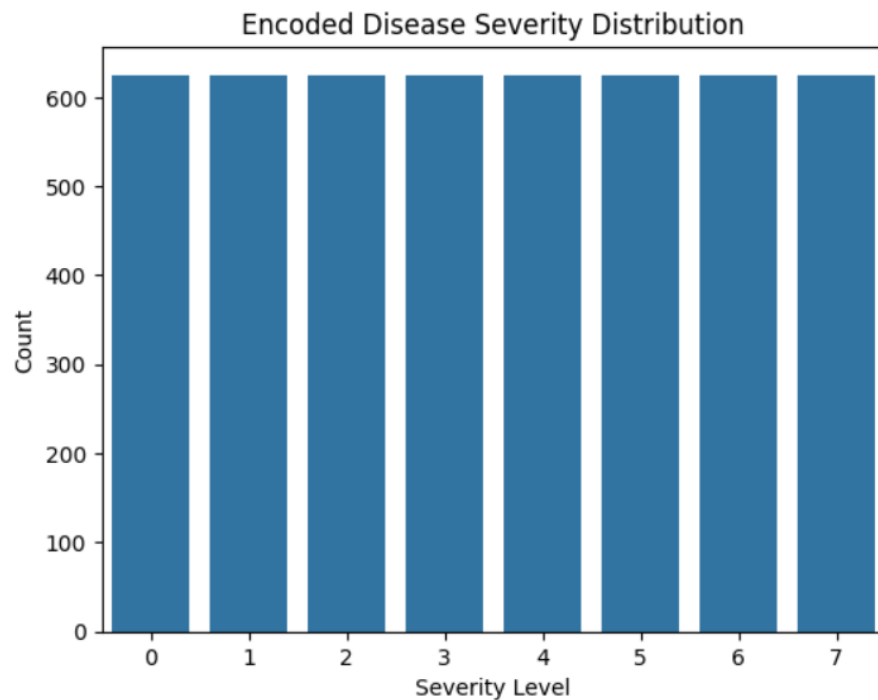


Figure 2: Box Plot of Features

The outliers and the distribution of medical indicators were identified by the box plot.

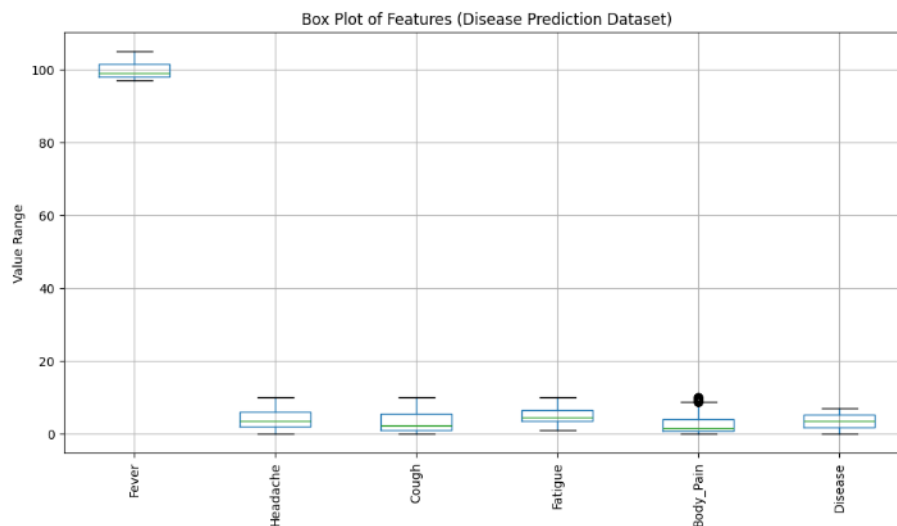
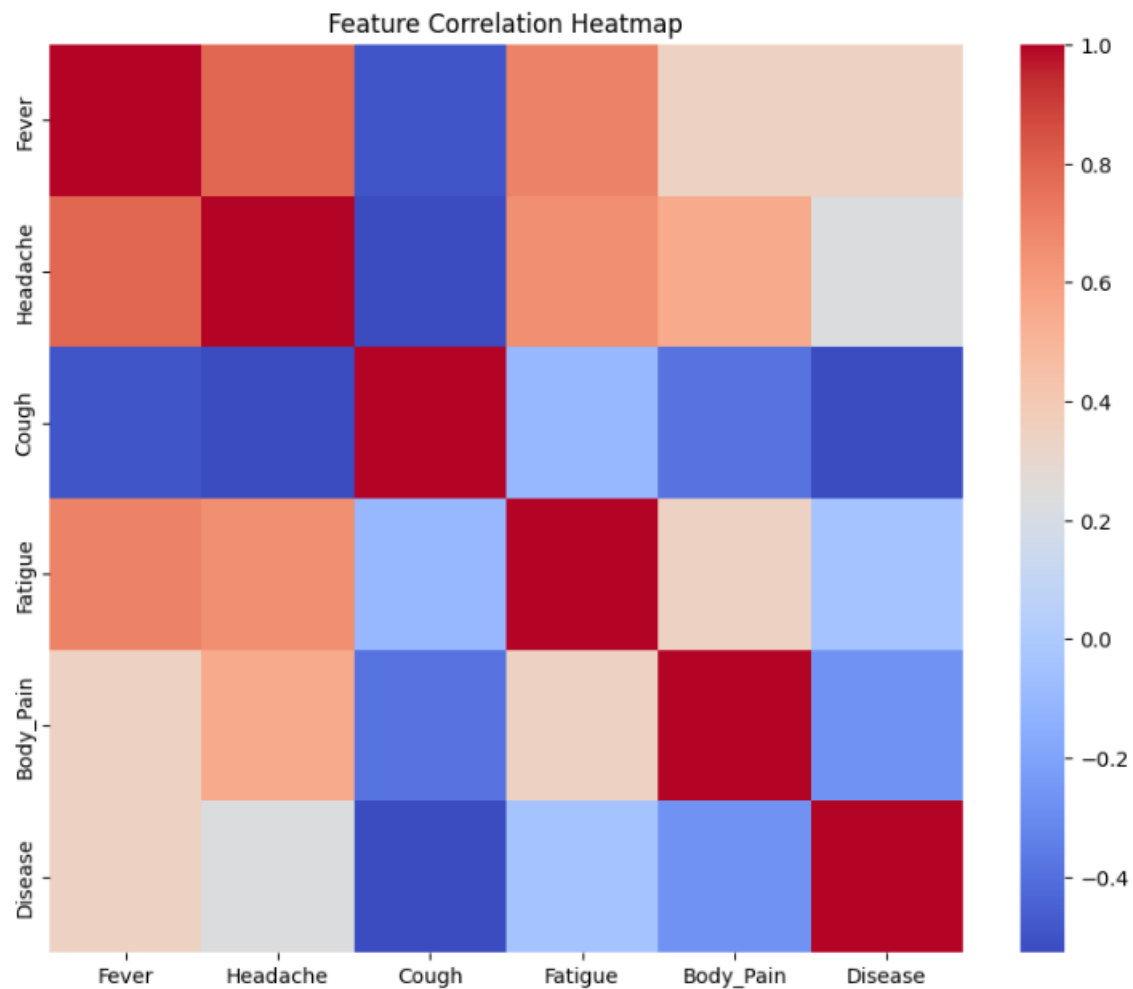


Figure 3: Correlation Heatmap

The heatmap is used to visualize the correlation of numerical features and the severity of the disease and it is used to identify powerful attributes.



## Methodology

### Data Preprocessing:

Preprocessing of data involved dealing with missing data, codification of the target variable and scaling of the numeric variables. The feature scaling had been used to make it compatible with the models that are sensitive to the magnitude of features like in neural networks.

The data was divided into training and testing to provide the unbiased evaluation.

### Model Building:

Regression Model of Neural Networks.

Multi-Layer Perceptron (MLP) regressor was used that was hidden and ReLU activation. Backpropagation was used to train the network to reduce the error in predictions.

Classical Regression Models.

Two classical regressions were used:

- Linear Regression
- Random Forest Regressor

The models give a comparison between the linear and non-linear methods.

### Model Evaluation:

The performance of the model was checked with the help of:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-squared (R2)

Figure 4: Exemplary and Projected Values.

This plot would be a comparison of forecasted and real severity.

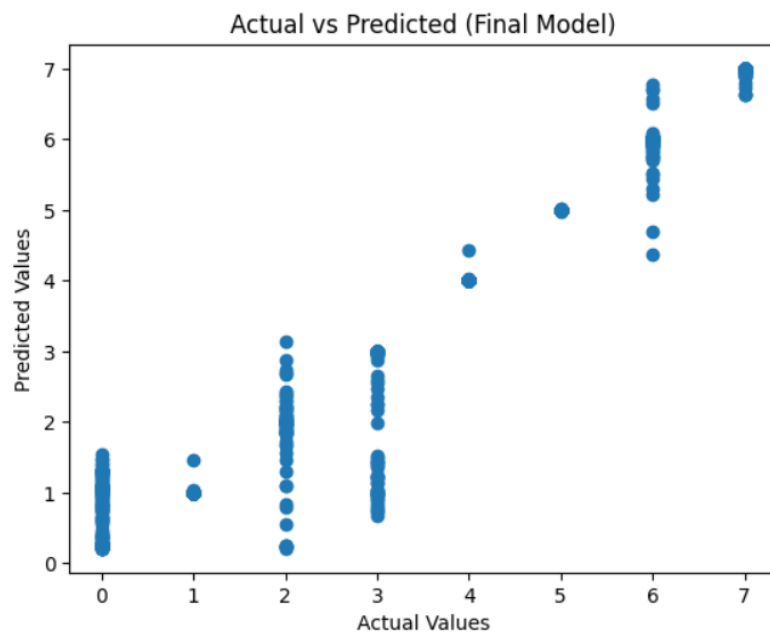
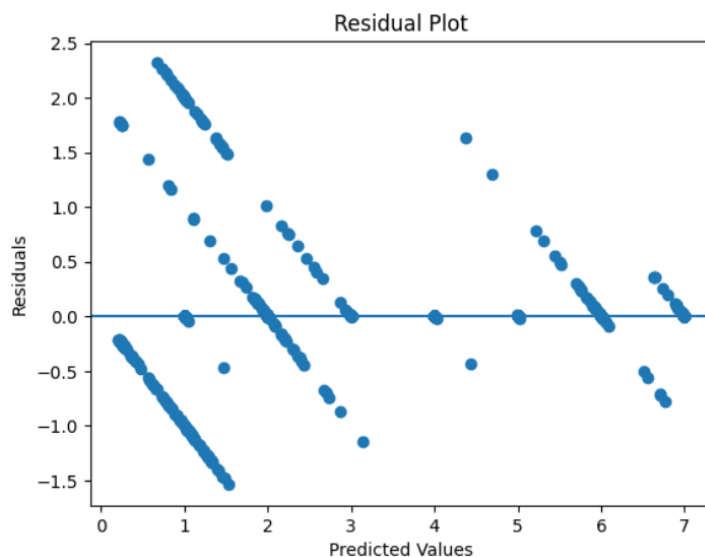


Figure 5: Residual Plot

Residual analysis was also employed to determine the distribution of error and stability of the model.





### Hyperparameter Optimization:

Cross-validation was done to introduce hyperparameter tuning in order to optimize the model performance and minimize overfitting.

### Feature Selection:

The review of feature selection methods was used to determine the most significant medical features. The dimensionality reduction enhanced the model generalization.

## Results & Conclusion

| Model            | CV Score | RMSE     | MAE      | R <sup>2</sup> |
|------------------|----------|----------|----------|----------------|
| Neural Network   | –        | 0.629985 | 0.380371 | 0.924254       |
| Ridge Regression | 0.590363 | 1.436230 | 1.168288 | 0.606317       |
| Random Forest    | 0.937635 | 0.515799 | 0.213085 | 0.949224       |

- Random Forest had best RMSE and R<sup>2</sup> compared to the other models.
- Neural Network was able to learn non-linear relationships but took more time to learn.
- Ridge Regression was more stable than the plain Linear Regression.
- Conclusion: the final model chosen is random forest because it has better predictive power and strength.

## Discussion

- Model Performance: random forest model always performed better as compared to other models.
- Effects of Hyperparameter Tuning: Higher Efficacy and lesser error of the Random Forest.
- Feature Selection: Better interpretability through specialization on most applicable symptoms.
- Limitations: Data set might not have all realistic variations in the course of the disease.
- Future Study: Discover ensemble models (XGBoost, Gradient Boosting), increase the size of the dataset, and use advanced feature engineering.

## References

- Kaggle: Predicting the disease in machine learning: Dataset (Kaushil268, 2020).
- Link: <https://www.kaggle.com/datasets/boffinbot/disease-prediction-dataset>
- Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, tensorflow.
- UN Sustainable Development Goals: Goal 3 - Good Health and Well-Being.